

Stage 3: Results and Analysis

Head-Level Characterization of the Single-hop Causal Heads in OLMo-2-7B

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1 Run and validation

Run `stage3_v1` completed on the cluster with three model replicas (six GPUs) and exit code 0. Every replica passed the gate before measurements began: behavioral replication drift 0, self-patch drift 0, zero change when corrupted activations are patched before the first differing token, reproduction of the saved Stage-2 `exact_delta` on the six probe heads, first-divergent-token alignment, QK/RoPE-aware reconstruction of the SDPA attention patterns, replication of the saved raw RI scores within the frozen tolerances, and agreement between the joint attention+value replacement and the ordinary head-output patch. The gate measured 0.151–0.152 s per patched forward and predicted 3.7 GPU-hours for the 263,328 new patched forwards (8,188 missing Scope-F patches, 176,820 position-scan patches, 74,760 attention/value factorial patches and 3,560 reverse patches).

The automatic CPU audit verified every checkpoint checksum, the coverage of all 105 heads on all 178 pairs, the position scan on the 40 common pairs, the 160 core prompts (20 common families $\times$ two orders $\times$ {base, corrupted, reorder, changed-query}), the 34,526 matched contextual-RI anchor comparisons, the weight diagnostic and both synthetic benchmarks, and wrote `summary.json` with `complete: true`. The 46 heads that lacked a Scope-F measurement now have one on all 178 pairs; no saved Stage-1/2 value was recomputed or replaced. All results are descriptive. Causal measurements (Scope P, Scope F, reverse patches, the attention–value factorial) are family-level means over the 89 discovery families with the two orders averaged within family; the position scan, attention and output profiles and the attended-name classification use the 20 common families (the latter counts eligible events, not families); the synthetic fingerprints average over trials. The 87 validation families remain untouched.

What Stage 3 measured. For the fixed inventory of 105 heads (48 RI-selected small-effect, 12 strong positive outside RI, 5 strong positive RI-selected, 8 strong negative, 7 moderate, 25 random controls) we report: (i) Scope-P vs. Scope-F importance with the clean/corrupted logit decomposition; (ii) the causal position profile (single-position patches at every test-block position, 40 common pairs) and reverse patching for the ten largest heads; (iii) the unconditional attention profile at the final colon and the changed-query, reorder and corruption controls; (iv) the contextual output projection $o_j^h W_U$ against the raw-embedding projection used by RI; and the four planned diagnostics: (v) weight copying scores, (vi) attended-name mover classification, (vii) matched contextual RI, (viii) the attention–value factorial, and (ix) two synthetic fingerprints. The completed optional controlled readout for L17H1 and L15H25 is reported in Section 7.7. Section 8 gives the head-profile table; Section 9 the evidence assessment of the three Stage-2 hypotheses; Section 10 the hand-off to Stage 4.

2 Notation

As in Stage 2, $M(x) = \text{logit}(y_c) - \text{logit}(y_r)$ at the first divergent answer token, with the sign fixed to the clean answer. For head h , $I_h = -[M(x_c; a_h \leftarrow a_h^r) - M(x_c)]$ with the corrupted donor

inserted at all original prompt positions (Scope P) or at the final colon only (Scope F, I_h^F). Positive importance means the patch moves the metric toward the corrupted answer. For every Scope-F patch the run stored the clean- and corrupted-answer logits, so $I_h^F = P_h + S_h$ where P_h is the rise of the corrupted-answer logit (“promotion”) and S_h the fall of the clean-answer logit (“suppression”). Reverse recovery is $M(x_r; a_h \leftarrow a_h^c) - M(x_r)$ (clean donor into the corrupted prompt). The test block reads “⟨mother⟩ is the mother of ⟨child⟩.” four times, then “Question: Who is the mother of ⟨child⟩? Answer:”. The corruption swaps the two answer-side mothers; the *query fact* is the fact whose child is asked about, and its mother is the answer.

3 Where the effect enters: Scope P vs. Scope F

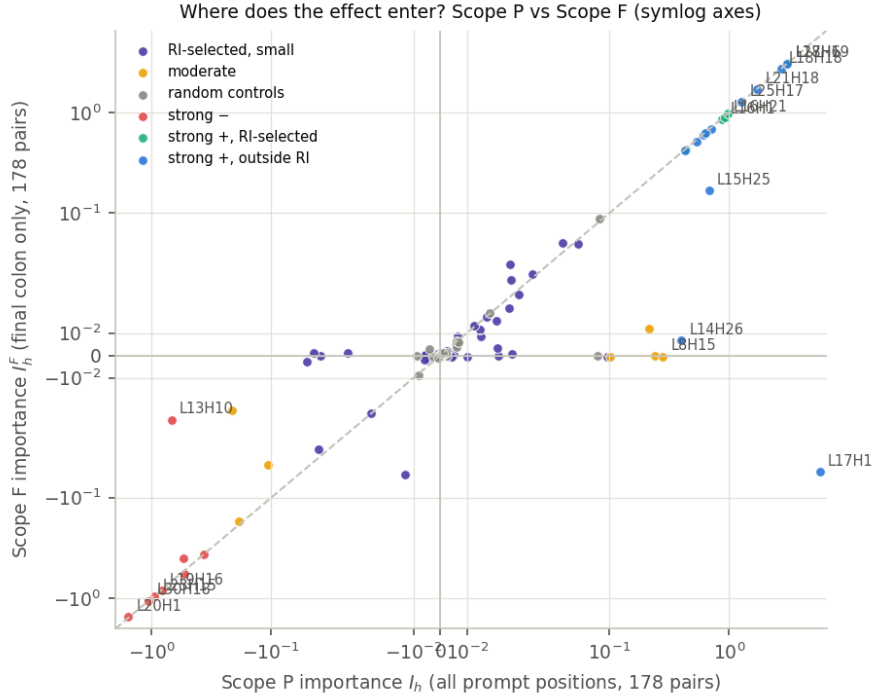


Figure 1: Scope-P against Scope-F importance for all 105 inventory heads (178 pairs, symlog axes). Heads on the diagonal deliver their whole effect at the answer position; heads on the horizontal axis act only at earlier positions.

Figure 1 completes the picture that Stage 2 could only sketch for 69 heads. Of the 25 heads with $|I_h| \geq 0.3$, 20 have I_h^F/I_h between 0.96 and 1.07: the final colon carries their entire effect, and the sign is consistent across families (F positive in $\geq 97.8\%$ of families for every strong positive head, in $\leq 4.5\%$ for every strong negative head). Five heads are different:

Head	I_h (P)	I_h^F	F/P	Position of the effect (Section 4)
L17H1	+5.701	−0.056	−0.01	query-fact sentence: mother tokens, “is”, period
L15H25	+0.679	+0.169	0.25	last token of the query-fact child
L14H26	+0.399	+0.007	0.02	diffuse; template words
L13H10	−0.681	−0.029	0.04	query-fact sentence: “is”, “of”, mother last token
L17H3	−0.546	−0.402	0.74	mixed: colon plus earlier positions

The most important single head of the whole project, L17H1 (5.7 logits, half the clean–corrupted gap), has *no* effect at the answer position. Stage 2 had labelled the layer-16–26 heads “readers at the

colon” from 69 heads; L17H1 was not among them, and the label does not hold for it. Twenty-five random controls have $|I_h^F| \leq 0.019$ except L18H22 (+0.089), which also has a Scope-P effect of the same size.

Logit decomposition. For every strong head the Scope-F effect is dominated by the corrupted-answer logit: P_h accounts for 93% (L18H19: +2.88 of +3.08), 96% (L18H18), 88% (L21H18), 93% (L22H5) and 98% (L20H1: -1.49 of -1.53; L23H15: -0.94 of -0.95; L19H16: -0.83 of -0.83). L27H6 is the exception: $P = +1.96$, $S = +1.10$; it both promotes the donor’s name and lowers the other. The negative heads therefore act by *lowering* the corrupted-answer logit when their corrupted-run activation is inserted, with almost no change of the clean-answer logit. Under the sign convention of Stage 2, a head that suppresses the name it retrieves on its own run produces exactly this pattern (the corrupted-run activation suppresses the corrupted answer); Section 7.3 tests that reading directly.

Reverse patching. For the ten fixed heads, inserting the clean activation into the corrupted prompt recovers the metric by almost exactly the noising effect (L17H1: +5.21 vs. 5.70; L18H19 +3.22 vs. 3.07; L27H6 +3.06; L18H18 +2.85; L21H18 +1.75; L25H17 +1.27; L22H5 +1.01; L20H1 -1.63; L30H18 -1.20; L23H15 -0.93). Recovery is symmetric with noising for every head, at Scope P and at Scope F alike (L17H1: -0.055 at the colon). This is conditional on the rest of the corrupted model and does not show that any head suffices alone.

4 Causal position profile

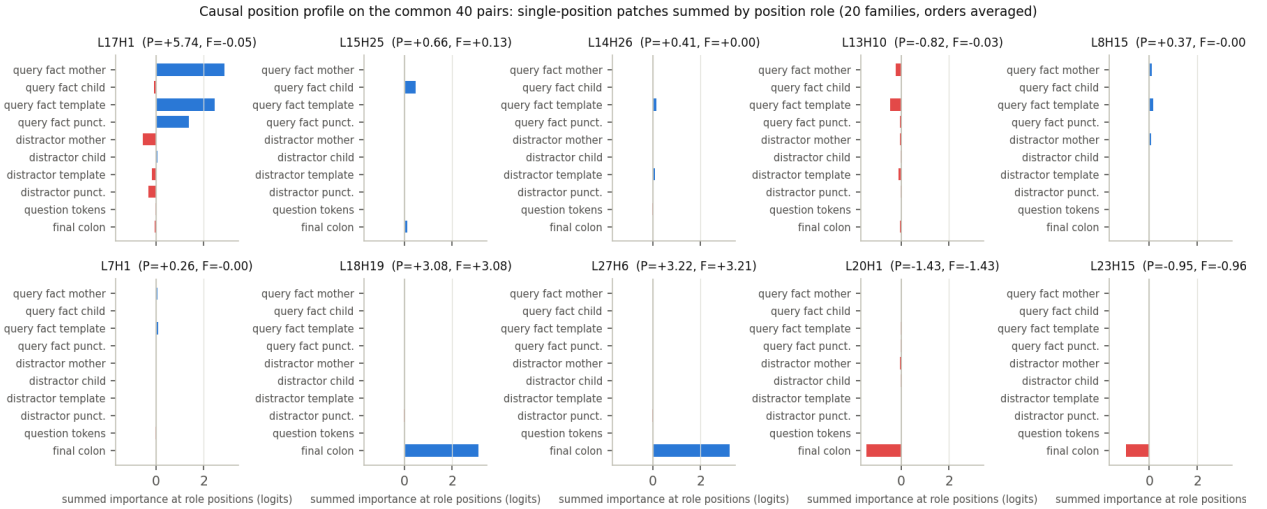


Figure 2: Single-position patches on the common 40 pairs, summed by position role within each prompt and averaged over the 20 families. Top: the four early-position heads and two moderate heads; bottom: four answer-position heads. Titles give Scope P and F on the same 40 pairs.

Sums of single-position effects vs. the all-position effect. Summing a head’s single-position effects over all test positions tracks its all-position Scope-P effect on the same 40 pairs in the aggregate: Spearman 0.913 over the 105 heads, and a signed mean residual of -0.135 logits for L17H1 (5.61 vs. 5.74). The signed mean hides large cancelling errors, however: for L17H1 the mean absolute residual per pair is 0.94 logits (per family, orders averaged, 0.89) and the maximum is 2.85 (2.33 per family). For the answer-position heads the residuals are small at every level (L18H19: mean absolute family residual 0.06, maximum 0.14; L27H6: 0.03, 0.09), for the trivial reason that all of their effect sits at one position. The single-position profile therefore locates *where* a head’s activity influences the answer; it is not a proven additive partition of the total effect, and within-head interactions between positions are not established to be small for L17H1.

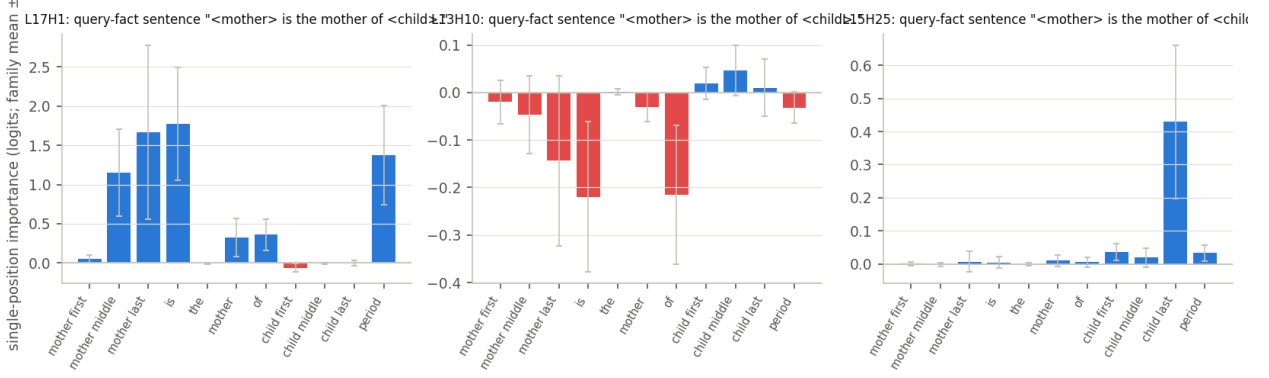


Figure 3: Token-by-token single-position importance inside the query-fact sentence (family mean $\pm$ SD, 20 families) for L17H1, L13H10 and L15H25.

L17H1 acts inside the query-fact sentence, after the mother’s first token. Figure 3 shows the profile: mother first token +0.05, middle +1.15, last +1.67, “is” +1.77, “the” 0.00, “mother” +0.33, “of” +0.36, child tokens $\approx$ 0, period +1.38. Nothing at the question, nothing at the colon. At the mother tokens the clean and corrupted *inputs* differ, so a head that writes a representation of the current name would show an effect there for trivial reasons; the effect at “is”, “of” and the period, whose inputs are identical in the two prompts, shows that L17H1’s output at those positions depends on the preceding mother and matters for the answer. The natural reading is that it carries mother-dependent information forward through the sentence into positions later heads read from; what that information is (the name, the relation, or a binding of the two) is not identified by this measurement (see Section 7.7 for the completed controlled readout). The same pattern with opposite sign describes L13H10 (“is” -0.22 , “of” -0.22 , mother last -0.14). At the distractor facts L17H1 has small *negative* effects (distractor mother -0.53 summed), as expected if the corrupted donor writes the wrong name identity into the other candidate fact.

L15H25 acts at the query child. Its effect is concentrated on the last token of the query-fact child (+0.43 of 0.66), with +0.13 at the colon. The child’s tokens are identical in the two prompts; the head therefore writes, at the child’s position, something that depends on the preceding mother. L14H26 (0.40) is diffuse over template words with no dominant site.

Answer-position heads. L18H19, L27H6, L18H18, L21H18, L25H17, L22H5, L16H21, L16H1, L21H6, L30H13, L21H23, L17H17, L17H24, L19H22 and all eight strong negative heads except L13H10 (and partly L17H3, $F/P = 0.74$) have flat profiles with a single bar at the colon.

5 Relation-tracking attention profile

The answer-position heads fall into three attention types (Figure 4; random controls: mean mass 0.017 on the query mother, maximum 0.109, 0.009 on the query child).

Mother-attenders. L27H6 (0.58 on the query mother), L21H18 (0.60), L23H15 (0.51), L22H5 (0.34), L19H22 (0.31), L21H6 (0.30), L19H16 (0.21), L26H23 (0.21), L20H1 (0.17), L18H18 (0.14), L18H19 (0.13), L17H24 (0.13). They attend to the answer’s own mention. Both signs occur: L23H15, L19H16, L26H23 and L20H1 attend to the same place as L27H6 and L21H18 but have negative effects.

Child-attenders. L16H21 (0.40 on the query child, 0.01 on the mother), L16H1 (0.36), L16H24 (0.37, small effect), L16H31 (0.14, negative). They attend to the second mention of the queried child inside the facts, not to the answer.

Self-attenders. L25H17 (0.57 on the colon itself, 1.27 logits), L13H10 (0.42), L21H23 (0.20), L30H13 (0.19), L17H17 (0.19). Self attention is their largest single target, but it is not exclusive: the eight fact-name spans together receive 0.03 (L25H17, L13H10), 0.09 (L30H13), 0.15 (L21H23) and 0.22

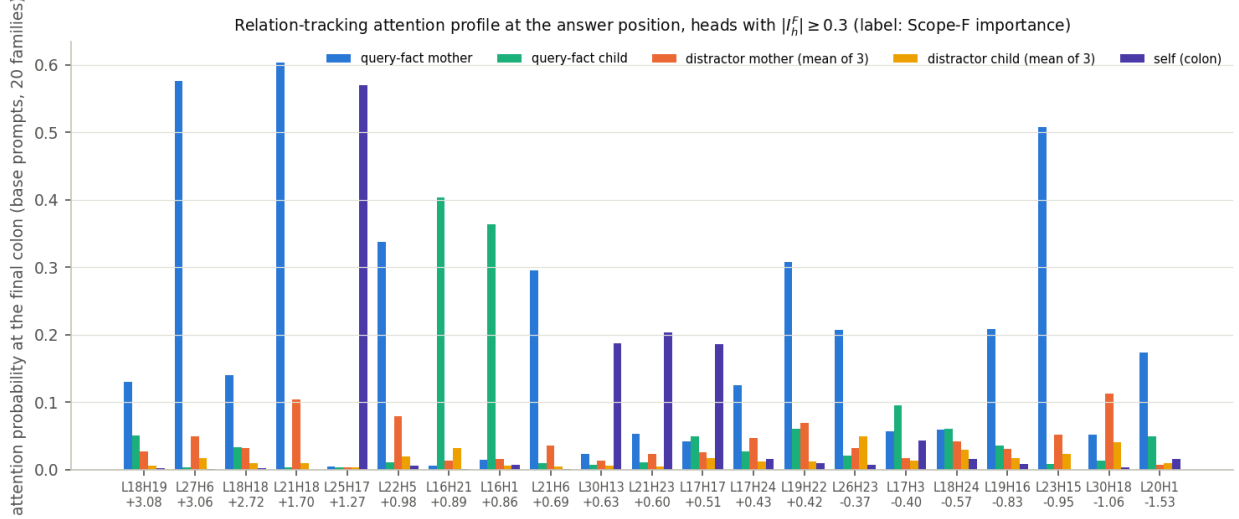


Figure 4: Attention mass at the final colon on base prompts (20 common families, both orders), for the 22 heads with $|I_h^F| \geq 0.3$.

(L17H17) of their mass, and the remainder is spread over other positions. For L25H17 most of the value it carries comes from the colon’s own residual stream; for L17H17 and L21H23 a fact-reading path cannot be excluded.

L30H18 (−1.06) is the one strong head that attends more to the *distractor* mothers (0.11 per fact) than to the query mother (0.05).

Attention follows the question, not the location or the token (Figure 5). Changing only the queried child moves the mother-attenders’ mass to the newly relevant fact in 100% of families (L21H18 +1.07, L23H15 +0.91, L27H6 +0.84, L19H22 +0.56, L21H6 +0.50, L22H5 +0.49, L26H23 +0.37, L19H16 +0.33, L20H1 +0.32, L18H19 +0.23, L18H18 +0.20), and moves the child-attenders’ mass to the new child (L16H21 +0.83, L16H1 +0.68, L16H31 +0.27). Random controls: at most +0.18. (The preference shift is a difference of two attention contrasts, so values above 1 are possible.) Reordering the facts and corrupting the mother change the family-mean attention on the query mother by at most 0.11 (L21H6) and 0.085 (L27H6) respectively, but these signed means also hide prompt-level variation: for L21H18 the mean absolute change under reordering is 0.15 per prompt (maximum 0.49) and under corruption 0.12; for L27H6 0.09 and 0.29; for L23H15 0.12 and 0.14 (the child-attender L16H21: 0.004 and 0.004). The heads therefore keep attending to the relevant fact *on average* when it is moved or its name is swapped, but positional invariance at the level of individual prompts is not established; the changed-query result is the stronger evidence, because it is consistent in sign in every family. The model itself answers the changed query correctly in 38 of the 40 changed-query prompts (mean margin 5.25 vs. 5.85 on the base prompts). L30H18 is again the exception: its mass moves *away* from the newly queried fact (−0.19, negative in every family). None of this establishes what the heads compute; it establishes that their routing is query-selective on these 20 families.

6 Contextual output profile

The raw-embedding projection is uninformative at the colon for every head (all values within ± 0.008 ; Spearman with I_h^F −0.07), as the algebraic argument of Stage 1 predicted. The contextual projection is informative but partial (Spearman +0.32). Three groups emerge.

Direct writers of the answer. L27H6 writes +2.99 logits for the query mother against the distractor mothers (first token; at the last-token anchor only +0.06); L30H13 +0.76; L22H5 +0.41; L21H6

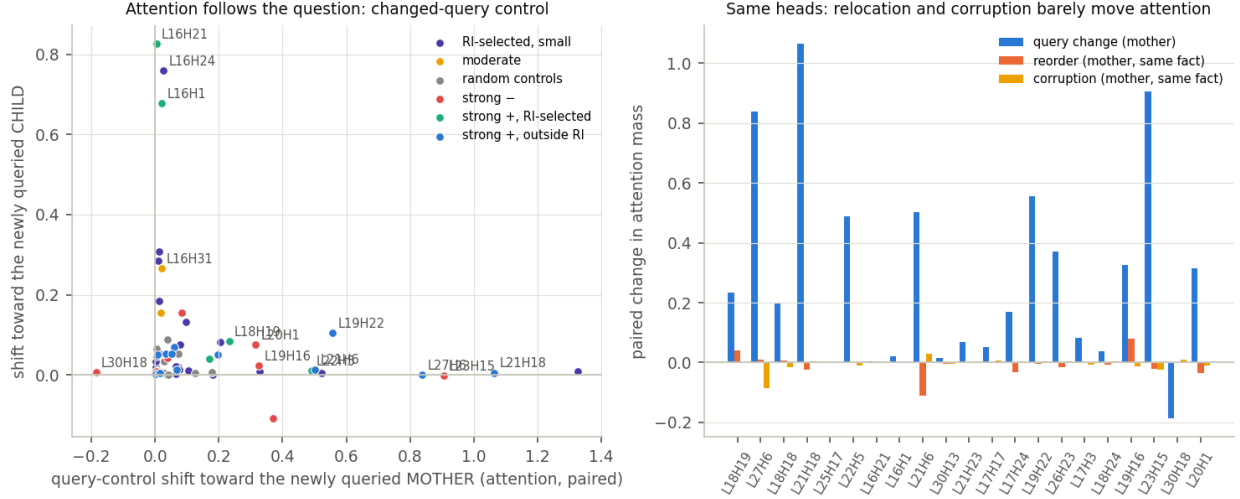


Figure 5: Left: paired attention shift when the question is changed to the other answer-side child (facts unchanged), toward the newly queried mother (x) and child (y), family means. Right: the same shift beside the paired change under reordering and under corruption for the strong heads.

+0.33; L21H18 +0.25. For L27H6 the projection is of the same size as its causal effect, but the two numbers are not an attribution equality: the projection is one clean vector through W_U before the shared post-attention norm, on 20 families, contrasting the query mother with the distractor mothers, whereas I_h^F is a clean-to-donor change in the full model on 89 families, at the first divergent answer token. L30H18 writes -0.88 : its output lowers the query mother relative to the distractors, consistent with its distractor-directed attention and its causal sign.

Heads whose projection does not explain their effect. L18H19 ($I^F = 3.08$) writes only +0.16 for the answer, L18H18 (2.72) +0.11, L20H1 (-1.53) -0.09 , L23H15 (-0.95) +0.04, L19H16 (-0.83) -0.04 , L25H17 (1.27) +0.16; L16H21 and L16H1 have contrasts within ± 0.02 on both mother and child tokens against 0.9-logit effects. For these heads the selected vocabulary projection of the clean output vector does not account for the measured intervention effect. The hypothesis this suggests is mediation by later components at the colon (later layers, since the colon is the last prompt token), but the present data do not establish it: the projection ignores the shared post-attention normalization, whose response to the donor vector is itself a candidate explanation, and it is a property of one clean vector rather than of the clean-minus-donor difference that the intervention inserts. A matched clean-minus-donor projection under an explicit normalization reference would be a sharper diagnostic; identifying a downstream path requires the Stage-4 path interventions.

All values are pre-normalization diagnostics of one head’s vector; they are not logit attributions, and the shared post-attention norm of OLMo-2 is not assigned to heads.

7 Planned diagnostics

7.1 Copying and token preference (weights)

The static copying score separates the sign of the effect but not its size: Spearman with signed I_h is +0.37 ($n = 105$), with $|I_h|$ +0.02. Positive copiers with positive effects: L21H6 (+0.35; the token is its own top output for 60% of names), L17H24 (+0.19), L19H22 (+0.17), L18H19 (+0.13), L21H18 (+0.12). Anti-copiers with negative effects: L26H23 (-0.20), L20H1 (-0.17), L19H16 (-0.09). Random controls lie in $[-0.044, +0.044]$. But the two largest heads at the colon are not copiers by weight (L27H6 +0.04, L18H18 -0.08), L17H1 is neutral (+0.01), and the strongest name-copier in the inventory, L30H18 (+0.60, far outside every other head), has a *negative* effect because it applies

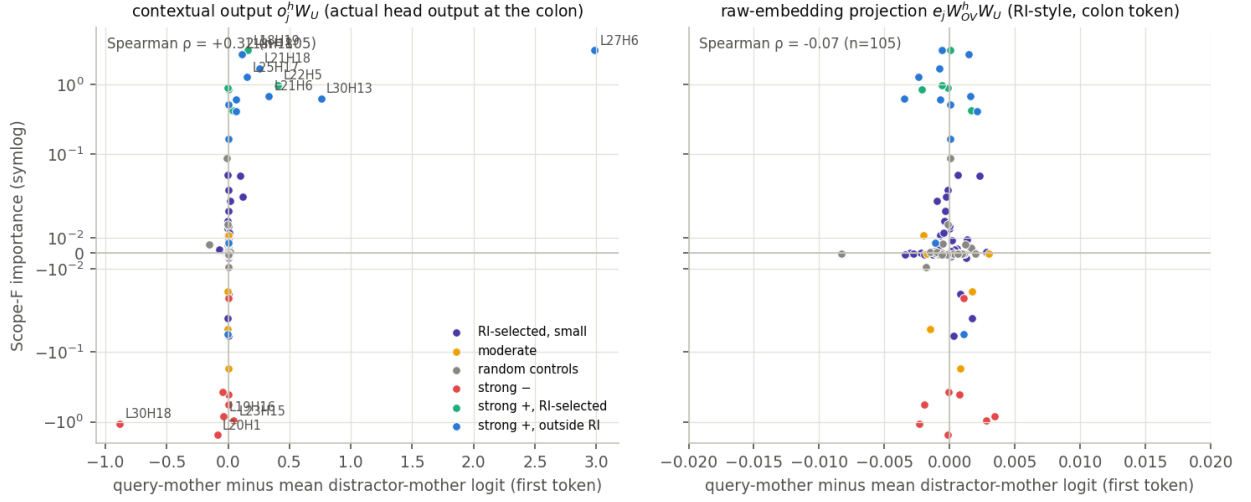


Figure 6: Scope-F importance against the vocabulary projection of the head’s output at the colon (query-mother first token minus mean distractor-mother first token). Left: actual contextual output $o_j^h W_U$. Right: the raw-embedding projection of the colon token used by RI, on the same x range as the left panel would need (± 0.02).

its copying to the distractor it attends to. Name-token and ordinary-word scores are similar for most heads (Figure 7, right); L30H18 alone is name-specific (+0.60 vs. +0.03). Weight alignment therefore neither identifies the important heads nor decides the copy-penalizing hypothesis on its own; the event-based test follows.

7.2 Copying and token preference (saved RI events by current-token role)

The plan’s first copying diagnostic partitions the saved Stage-1 test-only RI events by whether the *current* token belongs to the target name, to a control name, or to neither, and reports the raw target-minus-control gap in each partition. It was computed during preparation (`current_name_diagnostics.json`) and is reported here; it is distinct from the attended-name test, which concerns the token a head attends to.

For the positive movers with support in both partitions the name gap is clearly negative when the current token is a control name and near zero otherwise: L18H19 -0.023 (111 events) vs. $+0.000$ (51); L21H18 -0.017 (194) vs. $+0.001$ (185); L21H6 -0.025 (421) vs. -0.004 (116). For the anti-copying head L19H16 the control partition is positive ($+0.005$, 92 events) and the neither partition slightly negative. This is the pattern the copy-penalizing hypothesis predicts: a head whose OV circuit raises its current token’s logit scores badly on the name gap exactly at the events where the current token is a control name. The partitions differ in position and composition (control-token events are mostly self events on chain-middle names), so this is an association within the saved events, not an isolated causal explanation of the RI failure. Across the 33 heads with supported control-partition events the Spearman correlation of the control-partition gap with I_h is -0.26 ($p = 0.15$); the sample is too small for more than a direction. Many strong heads (L17H1, L27H6, L20H1, L30H18, L15H25, L21H23, L17H17) have no or almost no scored QK events and are insufficient in every partition.

7.3 Attended-name mover classification

Rule (fixed before results): a head is a *name mover* if the contrast is positive in $\geq 80\%$ of its attended-name events and a *negative name mover* if negative in $\geq 80\%$, with support ≥ 20 events in ≥ 10 families; below support the head is “insufficient”, never “neither”. Contrast = contextual logit of the attended name’s first token minus the mean over the other visible distinct names (both

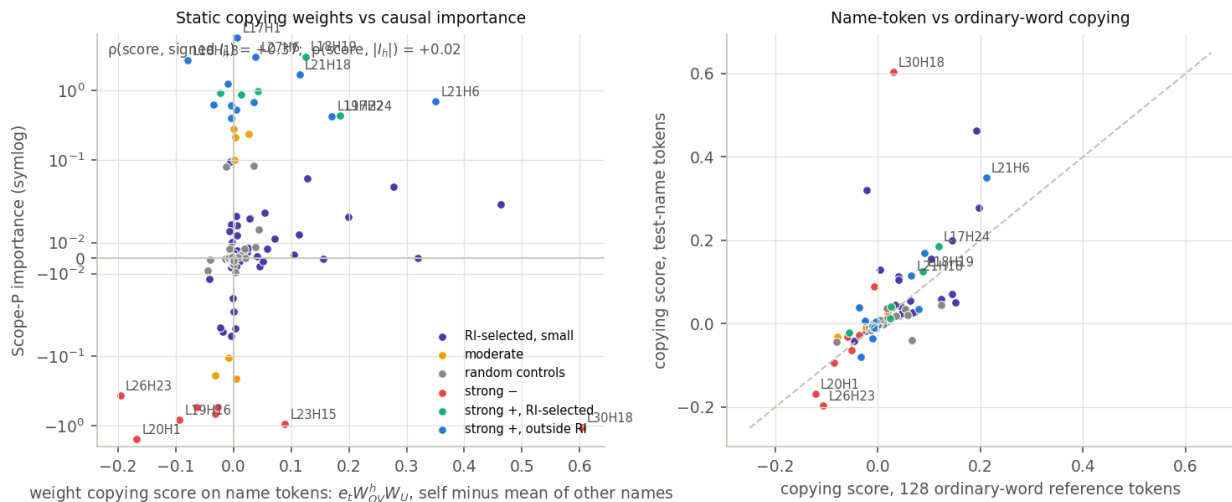


Figure 7: Left: weight copying score on the 314 first/last test-name token IDs ($e_t W_{OV}^h W_U$, self minus mean over other names) against signed Scope-P importance. Right: the same score on names against the 128-token ordinary-word reference sample.

roles); collisions of anchor token IDs are excluded. Sensitivity variants: last-token anchor, same-role controls only, non-self events only, colon site only, base prompts only. All 160 core prompts of all three replicas enter the classification (94,015 attended-name events over the 105 heads).

Results (Figure 8): six name movers, L27H6 (96% of 768 events, mean contrast +2.09), L17H24 (94% of 1,073), L26H31 (91% of 621, mean +2.09), L18H19 (84%), L22H5 (82%), L21H6 (82%); one negative name mover, L20H1 (17% positive, mean -0.10). Six of the seven labels are consistent with the sign of I_h^F ; the seventh, L26H31, is a strong mover of the name it attends to with a negligible individual causal effect ($I_h^F = +0.027$). Its attention rarely selects the answer: 12.9% of its 62 colon-site name events land on the answer (25% of the 16 base-prompt colon events, nine families), and it puts 0.09 of its colon mass on the query mother. Strong name movement with weak answer selection is a consistent descriptive account of its small effect; it is not established as the explanation, since redundancy with other movers and the specific intervention are alternatives. L21H18 (79%) and L19H22 (78%) miss the threshold and are name movers on colon-site events only; L18H18 (74%) is not labelled. One support detail matters for the strongest mover: L18H19’s overall label rests on 646 events, but its colon-site subset has only 17 events in nine families and is *insufficient* under the frozen rule, so the label describes its behaviour at earlier test positions, not specifically at the causal answer position. Every label disappears at the last-token anchor: what these heads move is the first token of the attended name, the token the model has to produce next.

The three pre-registered candidates give a mixed answer. L19H16 is negative in 77% of 1,906 events (just under the rule; negative name mover on the colon site only) and L26H23 in 79% (negative name mover with same-role controls). L23H15, however, favours the attended name in 64% of 2,428 events (mean +0.06) while its causal effect is -0.95 : it is *not* a negative name mover by its own output. Its attention is the same as L27H6’s (0.51 on the query mother, 100% query-tracking), it is the strongest induction/retrieval head of the inventory on the synthetic tasks (below), and its direct vocabulary write at the colon is +0.04. The “negative name mover” description inherited from Stage 2 is therefore not supported for L23H15; how its negative effect arises is open, and mediation by later components is the hypothesis to test in Stage 4. L30H18 is different again: its contrast on the attended name is positive (67%, mean +0.60) — it *promotes* what it attends to — and it attends to the distractor. Promotion of the distractor name is a hypothesis consistent with all three observations (attention, projection, causal sign); a targeted intervention on its attention would be needed to establish it. The label “negative mover” thus covers at least two different patterns; only L20H1 (and, marginally, L19H16 and L26H23) suppresses the name it attends to.

Head	I_h	Current token $\in$ target	Current token $\in$ control name	Neither
L18H19	+3.07	+0.1011 (1/1) [†]	−0.0228 (111/48)	+0.0003 (51/32)
L21H18	+1.70	−	−0.0168 (194/65)	+0.0011 (185/74)
L21H6	+0.70	−	−0.0245 (421/85)	−0.0044 (116/70)
L22H5	+0.98	−	−0.0187 (10/8) [†]	+0.0333 (14/5) [†]
L19H22	+0.43	−	−0.0148 (19/13) [†]	−0.0001 (79/47)
L16H21	+0.92	−	+0.0042 (149/67)	−0.0007 (307/82)
L16H1	+0.87	−0.0039 (88/34)	−0.0020 (242/75)	−0.0041 (331/82)
L25H17	+1.27	−	+0.0009 (73/32)	+0.0001 (58/25)
L14H26	+0.40	−	−0.0037 (27/13)	−0.0005 (9/8) [†]
L19H16	−0.82	+0.0031 (18/6) [†]	+0.0052 (92/59)	−0.0017 (144/61)
L23H15	−0.95	−	−0.0118 (30/26)	−0.0009 (32/25)
L26H23	−0.37	+0.0000 (1/1) [†]	+0.0092 (18/16) [†]	+0.0033 (39/22)
L13H10	−0.68	+0.0152 (2/2) [†]	−0.0157 (8/5) [†]	+0.0028 (31/19)

Table 1: Saved Stage-1 test-only RI events partitioned by the role of the *current* token (first anchor): family-weighted raw target-minus-control gap, with events/families. [†]: below the 20-event/10-family support rule. Heads with no scored QK events in a partition are marked −. Source: `results/stage3_inputs_v1/current_name_diagnostics.json`.

L17H1 favours the attended name in 77% of 3,469 events, but almost all are earlier- site events; its colon-site support is insufficient, consistent with its zero effect there.

7.4 Matched contextual RI

Replacing the raw current-token embedding by the actual normalized value input at the same position, with event IDs, controls, anchors and family weighting held fixed (70 heads with valid events; 27 inventory heads have no saved scored event and 8 have too few matched events), does not repair the Stage-1 statistics. On the main all-facts/all-test first-anchor name gap ($n = 68$ supported heads) the Spearman correlation with Scope-P importance moves from -0.26 (raw) to -0.17 (contextual); with the RI-supported family/order subset of pairs, from -0.28 to -0.19 . The raw target statistic is uncorrelated ($+0.01$ raw, $+0.02$ contextual); the last-anchor target statistic is negative under both (-0.31 , -0.23). The embedding choice is not what makes RI fail: RI scores the current position’s own residual, whereas the heads that matter act on what they attend to, or indirectly. This is a sensitivity analysis on a selected inventory, not an all-head result.

7.5 Attention–value factorial at the colon

For every head the value replacement alone carries the effect (L18H19: values $+3.16$, pattern $+0.08$; L27H6: $+2.59$ vs. $+0.27$; L18H18 $+2.88$ vs. $+0.11$; L21H18 $+1.70$ vs. $+0.01$; L25H17 $+1.29$ vs. -0.02). The routing term is small and non-zero only for the negative heads (L20H1 -0.22 , L30H18 -0.18 , L19H16 -0.16 , L17H3 -0.10 , L23H15 -0.09) and for L27H6 ($+0.27$). The family-mean interaction terms are also small (-0.27 for L18H18, $+0.21$ for L27H6, -0.16 for L18H19, otherwise near zero), but these means cancel large family-level interactions of both signs: the mean absolute interaction per family is 1.25 logits for L27H6 (maximum 4.58), 0.81 for L18H18 (maximum 3.27), 0.54 for L18H19 (maximum 3.25), 0.47 for L21H18 and 0.58 for L20H1, and the mean absolute routing term is 0.68 for L27H6 and 0.41 for L18H18. Value replacement carries the larger effect on average for the leading answer-position heads; a uniformly weak interaction between routing and values is not shown, and separability should not be assumed in Stage 4. On average the picture agrees with Section 5: the corruption barely moves the family-mean attention of these heads, so what changes is mostly the content at the attended position and that content is what the reader

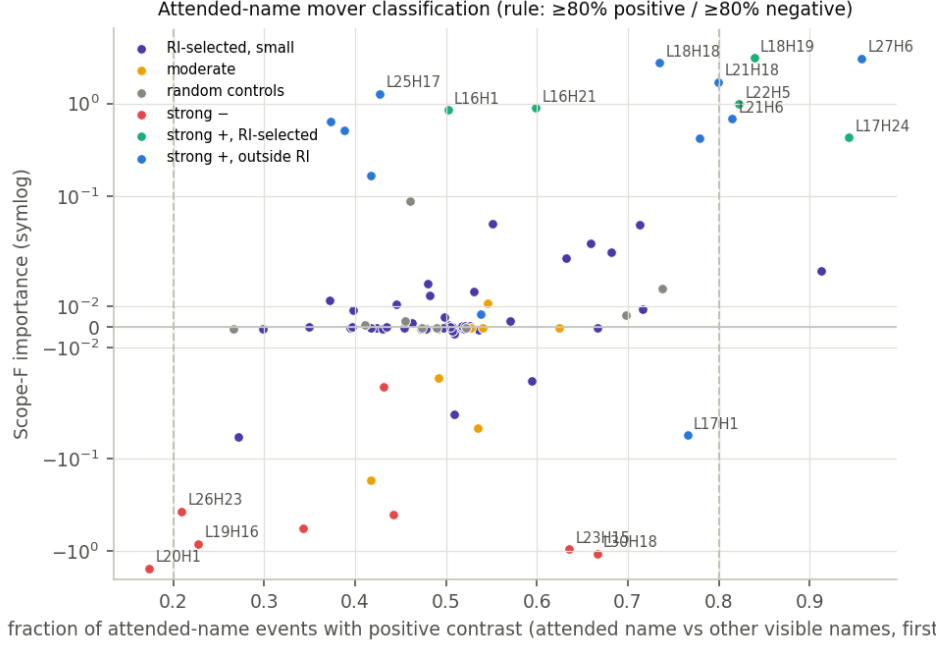


Figure 8: Fraction of attended-name events (test-block rows of the core prompts whose attention argmax lies inside a visible test name) in which the contextual output $o_j^h W_U$ favours the attended name’s first token over the other visible names, against Scope-F importance. Dashed lines: the fixed 80% thresholds.

carries to the colon. The factorial does not separate “name copying” from “reading a relational binding”: both predict value dominance.

7.6 Repeated-sequence and retrieval fingerprints

The model solves both tasks (key–value retrieval 100% of 32 trials; repeated sequence 96.1% of 992 probes), so both fingerprints are defined for every head. Retrieval-type attention (argmax on the copy source) is shown by L19H16 (0.97 of probes), L21H18 (0.88), L23H15 (0.84), L18H18 (0.56), L16H1 (0.47), L18H19 (0.19), L22H5 (0.16), L26H23 (0.13); induction-type attention on the repeated sequence by L23H15 (0.66), L21H6 (0.64), L16H1 (0.45), L21H18 (0.39), L19H16 (0.36), L22H5 (0.22), L26H23 (0.15). Random controls reach at most 0.44 and 0.17. Output sign on the gold token separates the two kinds of source-attender: L21H6 (+0.42 on repeats), L21H18 (+0.15/+0.09), L18H19, L22H5 and L16H1 promote the copied token; L19H16 (−0.24/−0.10), L26H23 (−0.19/−0.17) and L23H15 (−0.05/−0.02) attend to the source and *do not* promote it. L25H17, L30H13 and L21H23 have positive gold output with self-attention on both tasks, as in the kinship task. L27H6, L17H1, L18H18 (output), L20H1 and L30H18 show no fingerprint on either probe. Absence on these two probes is not evidence of task specificity; other synthetic tasks might reveal a transferable pattern. Transfer to a synthetic task supports a behavioural capability and does not show that the same algorithm solves the kinship task.

7.7 Controlled representation readout (Stage-3 plan, Section 1.5)

Question and completed run. The optional readout asks whether the causal sites of L17H1 and L15H25 expose information about the mother’s identity, and whether that readout depends on the selected head. Run `stage3_readout_v1` contains 1,920 records: the 40 common pairs (20 discovery families, both orders), four writer/site combinations, six conditions and two readout prompts. The frozen plan matches the manifest hash; the gate passed with zero identity-injection and self-patch

each family, then average the 20 family values. The measurement concerns first-token preference, not full-name generation. The source residual combines multiple components; head dependence is assessed only through the paired intervention.

Writer	Site	Readout	Clean C	Source swap ΔC	Head-site ΔC	Negative families
L17H1	is	identity	+0.51465	−0.52070	−0.45840	20/20
L17H1	period	identity	+0.28789	−0.04609	−0.04121	20/20
L17H1	is	mother_of	+0.25459	−0.00903	−0.00981	17/20
L17H1	period	mother_of	+0.26479	−0.02686	−0.02319	20/20
L15H25	child last	identity	+0.18213	+0.02021	+0.00605	8/20
L15H25	child last	mother_of	+0.24312	+0.00181	−0.00020	9/20
L15H25	is	identity	+0.29395	−0.15117	+0.00303	7/20
L15H25	is	mother_of	+0.25063	−0.00171	+0.00020	5/20

Table 2: Controlled readout, family-level descriptives. Source swap uses the intact corrupted prompt; head-site replaces only the selected head at the capture position. The last column counts families with a negative head-site change after averaging orders. The head-span condition is identical by construction at this capture layer (see below). Values come from `readout_family_summary.csv`.

L17H1: identity-sensitive information, strongest at “is”. In the identity readout, the contrast at “is” falls from +0.515 to +0.056 after head replacement, compared with −0.006 for the corrupted source and +0.002 for the other-fact control. The head-induced shift is negative in every family (median −0.459, range [−0.701, −0.234]). Its magnitude is about 88% of the mean full-source-swap shift. This supports a contribution of the head’s local output to the mother-token preference exposed by this readout. The ratio compares two readout changes; it is not a fraction of encoded information or of the original task’s causal effect.

At the period, the identity shift is weaker but also negative in every family: −0.0412 for head replacement versus −0.0461 for source swap, with median −0.0410 and range [−0.0723, −0.0117]. The `mother_of` readout at the period also responds consistently (−0.0232 versus −0.0269). These small responses do not by themselves establish a relational binding. For comparison, the original-task single-position effects are +1.769 logits at “is” and +1.377 at the period. The head’s approximately 5.7-logit Scope-P effect is an all-position measurement, not the effect at either site; none of these effects is numerically decomposed by the readout shifts.

L15H25: the child’s-site content remains unresolved. The child’s last-token site has original-task importance +0.429 logits, positive in every family. Replacing the head changes the captured residual (mean norm difference 1.106), but the `mother_of` contrast barely changes: −0.00020 under head replacement, and only +0.00181 under the full source swap. Identity readout also lacks a consistent mother-swap response there. Since even the source-swap control does not expose the expected information, the small head effect cannot distinguish missing information from an insensitive readout. The query mother ranks first among the candidates in 55% of `mother_of` probes for clean, head-replaced, corrupted and no-injection conditions alike. Nor does identity readout reliably recover the child: its first token ranks first in only 15% of intact probes.

At L15H25’s secondary “is” site, identity readout does respond to source swap (−0.151, negative in 18/20 families), while the head shift is approximately zero (+0.003), consistent with that site’s small original-task effect (+0.00394). This does not assign an address-only role to the head at its different, causal child-last site.

Readout quality and scope. For every intact-clean probe, the top next token is “hello” under identity and “Mary” under `mother_of`; the experiment has not freely decoded the names. At L17H1/“is”, identity readout assigns the query mother’s first token mean probability 0.000373 (0.0373%), compared with 0.0000429 without injection, 0.000282 for the other-fact residual, 0.000290

after head replacement and 0.000285 for the corrupted source. Thus there is a positive probability-relevance check and a source-specific contrast, but low absolute decoding probability. These are means of per-example probabilities, not exponentiated mean log probabilities. A null readout does not establish absent information; a `mother_of` prompt does not itself establish successful relation decoding.

Why head-site and head-span are identical. All 320 paired readout outputs are exactly identical between these conditions. The head is replaced after its layer’s attention mixing, and the residual is captured at the output of that same block. The remaining projection, normalization, residual addition and MLP act independently at each position. Extra replacements at earlier positions therefore cannot change the captured position within this block. This is a structural property of the intervention, not evidence that earlier writes are unimportant. Testing their propagation requires a later capture layer or a path intervention. The two conditions must not be counted as independent corroboration.

Implication. The extension strengthens the hypothesis that L17H1 contributes mother-identity-sensitive information at “is” and the period, while leaving the representation at L15H25’s child-last site unresolved. No writer-to-reader edge or circuit has been demonstrated. Stage 4 should test consumers of these actual source positions and retain L15H25 as a causal candidate despite the inconclusive readout.

8 Head-profile table

Table 3 summarizes the 25 strong heads; the per-head evidence cards (`analysis/head_cards.md`) and the full tables cover all 105. For every head and analysis the status is one of completed, insufficient support, unavailable input, or unresolved: contextual RI is insufficient for 35 heads (no or too few saved QK events), the conditional synthetic scores are complete for all heads, the attended-name label is insufficient for 19 heads (mostly heads whose attention never lands on a name), and every other cell is completed. The controlled representation readout of Stage-3 plan Section 1.5 is completed for L17H1 and L15H25 (Section 7.7). It provides an intervention-dependent mother-token preference for L17H1, strongest at “is”, while L15H25’s child-last representation remains unresolved under the current readouts. These measurements supplement the head-profile table. Broad spectral analysis and function-vector mediation were not run.

9 Evidence assessment of the Stage-2 hypotheses

Copying. Partly supported, with an important correction. Six heads (L27H6, L17H24, L26H31, L18H19, L22H5, L21H6) promote the first token of the name they attend to in $\geq 80\%$ of events, and L27H6 writes the answer directly (+2.99 logits). Weight copying scores have the right sign for these heads and for the anti-copyers L20H1, L26H23, L19H16. The Stage-2 conjecture that the negative name-gap correlation arises because “the causal heads copy names” is therefore consistent with the positive movers. But copying is neither necessary nor sufficient: L18H18 (2.7 logits) and L27H6 have weight scores near zero or negative, the strongest weight copier L30H18 hurts the answer by copying the distractor, L26H31 moves names as strongly as L27H6 but attends to the wrong ones and has no effect, and the largest head, L17H1, does not act at the colon at all.

Contextual information. Supported for the location, with limited new content evidence from the controlled readout. The heads’ effects come mainly through their values (factorial, on average), their attention tracks the queried fact under query change (and on average under relocation), and their raw-embedding projections are uninformative while the contextual projections are partly informative. Contextual RI at the current position does not help, because the information the readers use is at the attended position, written there by earlier layers (Section 4). The controlled readout (Section 7.7) supports a contribution of L17H1 to mother-identity-sensitive information at “is and the period; it

Head	Group	I_h (P)	I_h^F	Attention at colon	Ctx. out.	Copy	KV	Rep.	Attended-name label
		178 pairs	178 pairs	Q-mother/Q-child/D-mother/self	Δ mother	weights	src att.	src argmax	(fraction positive; events/families)
L17H1	S+	+5.70	-0.06	0.05/0.06/0.02/0.04	-0.00	+0.01	0.00	0.00	neither (77%; 3469/20)
L18H19	S+RI	+3.07	+3.08	0.13/0.05/0.03/0.00	+0.16	+0.12	0.16	0.02	NM (84%; 646/20)
L27H6	S+	+3.07	+3.06	0.58/0.00/0.05/0.00	+2.99	+0.04	0.05	0.01	NM (96%; 768/20)
L18H18	S+	+2.70	+2.72	0.14/0.03/0.03/0.00	+0.11	-0.08	0.27	0.06	neither (73%; 332/20)
L21H18	S+	+1.70	+1.70	0.60/0.00/0.10/0.00	+0.25	+0.11	0.46	0.39	neither (80%; 1656/20)
L25H17	S+	+1.27	+1.27	0.00/0.00/0.00/0.57	+0.15	-0.01	0.01	0.00	neither (43%; 625/20)
L22H5	S+RI	+0.98	+0.98	0.34/0.01/0.08/0.01	+0.41	+0.04	0.18	0.22	NM (82%; 730/20)
L16H21	S+RI	+0.92	+0.89	0.01/0.40/0.01/0.00	-0.01	-0.02	0.04	0.00	neither (60%; 551/20)
L16H1	S+RI	+0.87	+0.86	0.01/0.36/0.02/0.01	-0.00	+0.01	0.28	0.45	neither (50%; 669/20)
L21H6	S+	+0.70	+0.69	0.29/0.01/0.04/0.00	+0.33	+0.35	0.12	0.64	NM (81%; 858/20)
L15H25	S+	+0.68	+0.17	0.03/0.06/0.02/0.01	-0.00	+0.04	0.00	0.00	neither (42%; 1797/20)
L30H13	S+	+0.63	+0.63	0.02/0.01/0.01/0.19	+0.76	-0.03	0.03	0.00	neither (37%; 917/20)
L21H23	S+	+0.61	+0.60	0.05/0.01/0.02/0.20	+0.07	-0.00	0.04	0.00	insuff. (25%; 20/8)
L17H17	S+	+0.53	+0.51	0.04/0.05/0.03/0.19	+0.00	+0.01	0.01	0.00	neither (39%; 103/19)
L17H24	S+RI	+0.43	+0.43	0.13/0.03/0.05/0.02	+0.03	+0.18	0.02	0.00	NM (94%; 1073/20)
L19H22	S+	+0.43	+0.42	0.31/0.06/0.07/0.01	+0.07	+0.17	0.04	0.01	neither (78%; 1127/20)
L14H26	S+	+0.40	+0.01	0.01/0.05/0.00/0.01	+0.01	-0.00	0.00	0.00	neither (54%; 959/20)
L26H23	S-	-0.37	-0.37	0.21/0.02/0.03/0.01	-0.04	-0.20	0.16	0.15	neither (21%; 1670/20)
L18H24	S-	-0.53	-0.57	0.06/0.06/0.04/0.02	+0.00	-0.06	0.02	0.00	neither (34%; 70/15)
L17H3	S-	-0.55	-0.40	0.06/0.10/0.02/0.04	+0.00	-0.03	0.08	0.00	neither (44%; 319/20)
L13H10	S-	-0.68	-0.03	0.00/0.01/0.00/0.42	+0.00	-0.03	0.00	0.00	neither (43%; 700/20)
L19H16	S-	-0.82	-0.83	0.21/0.04/0.03/0.01	-0.04	-0.09	0.55	0.36	neither (23%; 1906/20)
L23H15	S-	-0.95	-0.95	0.51/0.01/0.05/0.00	+0.04	+0.09	0.49	0.66	neither (64%; 2428/20)
L30H18	S-	-1.06	-1.06	0.05/0.01/0.11/0.00	-0.88	+0.60	0.05	0.00	neither (67%; 2528/20)
L20H1	S-	-1.56	-1.53	0.17/0.05/0.01/0.02	-0.09	-0.17	0.02	0.00	neg. NM (17%; 437/20)

Table 3: Head-profile table for the 25 inventory heads with $|I_h| \geq 0.3$, ordered by Scope-P importance. Group codes: S+ strong positive outside RI; S+RI strong positive RI-selected; S- strong negative. Attention: mean probability mass at the final colon on base prompts (20 common families) on the query-fact mother, the query-fact child, one distractor mother (mean of three) and the colon itself. Ctx. out.: contextual output projection $o_j^h W_U$ at the colon, first token of the query mother minus mean of the distractor mothers. Copy: weight copying score on name tokens (self minus mean of other names). KV / Rep.: synthetic retrieval and repeated-sequence fingerprints (attention to the copy source; argmax on the source). Attended-name label under the fixed 80% rule (NM = name mover; Section 7.3), with the fraction of events favouring the attended name and the event/family support.

does not establish a relational binding, and the content at L15H25’s child-last site remains unresolved. How that information is used by downstream components is a Stage-4 question.

Early vs. late positions. Supported and sharpened. The effect of L17H1, L15H25, L14H26 and L13H10 is written inside the query-fact sentence (L17H1 at the mother tokens, “is” and the period; L15H25 at the child’s last token) and read later; all other strong heads act only at the colon. The earlier “layer-6–11 writers” of Stage 2 (L6H24, L7H1, L8H15, L9H16, L12H17) are confirmed to have zero colon effect, but their all-position effects are small (≤ 0.28); the important writer is L17H1, in the same layer band as the readers.

What Stage 3 does not establish. No edge between any writer and any reader has been tested; single-position additivity within one head says nothing about additivity across heads; the labels “name mover” and “negative name mover” describe the sign of one head’s output on its attended token and are not mechanism claims; the changed-query control is a discovery measurement on 20 families; and all thresholds are descriptive.

10 Hand-off to Stage 4

Stage 4 receives the following components and specific communication hypotheses.

1. **Writer L17H1 → mother-attending readers.** L17H1 writes at the query mother’s tokens, “is” and the period; L27H6, L21H18, L22H5, L18H19, L18H18 attend to the mother’s tokens at the colon. Path patching L17H1’s output at those positions into the readers’ keys vs. values will test whether L17H1 supplies the content the readers carry (values) or the address they select (keys). The value dominance of the factorial predicts values. The controlled readout strengthens the content hypothesis at “is and the period, but these positions are distinct from the mother-name tokens: test their actual consumers, including possible intermediate components, rather than assuming that mother-attending readers consume these sites.
2. **L15H25 → child-attenders.** L15H25 writes at the query child’s last token; L16H21 and L16H1 attend to the child at the colon and have no direct vocabulary output. Test whether they route through L15H25’s write and where their 0.9-logit effects surface (which later heads read them). Retain L15H25 despite the inconclusive child-last readout: the source-swap control did not expose the mother reliably, so a null head readout does not rule out a binding or other useful content.
3. **Heads whose projection does not explain their effect.** L18H19, L18H18, L20H1, L23H15, L19H16 have colon effects far larger than their vocabulary projections. Test the mediation hypothesis: path-patch their output into later heads and MLPs at the colon, and compute a matched clean-minus-donor projection under an explicit normalization reference; L23H15’s and L30H18’s negative effects in particular need this before any type label. L17H3 ($F/P = 0.74$) is a mixed-position candidate whose earlier sites should be located first.
4. **Positive vs. negative movers.** Joint interventions on {L27H6, L21H18, L22H5, L21H6} against {L20H1, L19H16, L26H23} to test redundancy and whether the negative heads calibrate the positive ones; group ablation before any faithfulness claim.
5. **Validation.** Only after edges are established: minimal-circuit faithfulness on the 87 held-out families and the reorder/changed-query conditions.

11 Conclusion

Stage 3 replaces the Stage-2 sketch with a measured profile of every inventory head. The largest causal head, L17H1, is not a reader at the answer position: it acts inside the query-fact sentence, at

and after the mother’s name, and its effect is recovered symmetrically by reverse patching. Twenty strong heads deliver their whole effect at the colon, mainly through their values, with attention that follows the queried fact under a changed query and, on average, under relocation and name swapping. Among them, six promote the first token of the name they attend to (L27H6 directly writing the answer at +3 logits; L26H31 as strongly, but at the wrong names and without effect), one suppresses it (L20H1), several have effects that their own vocabulary projection does not explain (L18H19, L18H18, L23H15, L19H16), and one attends to the distractor and promotes it, consistent with its negative effect (L30H18). Neither the raw-embedding RI projection nor its contextual variant tracks these effects, because the relevant information sits at the attended positions and, for the indirect heads, downstream. Stage 3 therefore ends with explicit components and edges to test, not with a circuit. The optional readout adds a consistent head-dependent mother-token preference for L17H1, strongest at “is, while leaving L15H25’s child-last representation unresolved. These are controlled readout findings rather than successful free-text name decoding or evidence of a writer-to-reader channel.